Assignment 8

Due Sunday November 9th, 2025, Self Grading Due Sunday November 16th, 2025

1. Review Chapter 5 and start reading Chapter 6.
2. Madhow Problem 4.16
3. Gaussian Random Variables
   (a) Show that the mean and variance of a Gaussian random variable X with the density function give by
   $$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}$$
   are μ_X and σ_X^2, respectively.
   (b) Show that for a Gaussian random variable X with mean μ_X and variance σ_X^2, the transformation $Y = (X - \mu_X)/\sigma_X$ converts X to a normalized Gaussian random variable with zero mean and unit variance.
4. Probability of Bit Error with PAM
   In a pulse-amplitude modulation (PAM) transmission scheme, binary data are represented with voltage levels of $+A$ for a 1 and $-A$ for a 0. The received data can be represented by the random variable Y, defined as
   $$Y = X + N,$$
   where N is a Gaussian random variable with zero mean and variance σ^2. A 1 is transmitted 75% of the time, and a 0 is transmitted 25% of the time.
   (a) Using Maximum A Posteriori (MAP) Estimation, derive the decision rule.
   (b) Express the probability of error.
5. **Self-grade Homework 7.** Fill out the self-grade rubric for Homework 7. Attach to Homework 8 submission.